

OMIS 105

Database Management Systems

Week 1 — Foundations

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Agenda (Today)

- Why databases matter
- What is data?
- File systems vs DBMS
- Relational thinking (preview)
- First SQL queries
- Hands-on practice

Why Should You Care?

Think about apps you use daily:

- Banking apps
- Amazon
- Netflix
- Uber

👉 All powered by databases

Real-World Example

Imagine Amazon without a database:

- Orders lost ✗
- Prices inconsistent ✗
- Inventory incorrect ✗

👉 Databases prevent chaos

What is Data?

Data = raw facts

Examples:

- Name: "Alice"
- Price: 1000
- Date: 2026-01-01

What is Metadata?

Metadata in database tables is "**data about data**"
—the underlying blueprint that defines how a table is structured, organized, and governed rather than the actual records stored inside it.

```
-- metadata
CREATE TABLE employees (
  name  VARCHAR,
  age   int,
  salary int
);
```

```
-- data
INSERT INTO employees
VALUES
('alex', 25, 78000);
```

From Data to Information

Data → processed → Information

Example:

- **Raw Data:** sales transactions
- **Insight:** "Top-selling products in CA"
- **Insight:** "Top-5 Customers in NY"
- **Insight:** "Total sales of iPhone 16 in December 2025"

What is a Database?

A structured collection of data

Key properties:

- Organized
- Persistent
- Queryable

File System vs Database

File System (Excel / CSV)

- No relationships
- Hard to maintain
- Data duplication

Database

- Structured
- Connected data
- Efficient queries

Problem: Data Duplication

customer	order	price
Alice	Laptop	1800
Alice	Phone	1200
Alice	Phone	1100
Jane	Laptop	1900
Jane	Phone	1400

👉 What if name changes?

Solution: Database Design

Separate tables:

- Customers
- Orders

👉 One change → consistent everywhere

What is a DBMS?

Database Management System

Responsibilities:

- Store data
- Retrieve data
- Ensure consistency
- Handle multiple users

Examples of DBMS

- DuckDB
- MySQL
- PostgreSQL
- Snowflake
- Oracle

Roles in Database World

- Developer → builds apps
- Analyst → queries data
- DBA → manages database

Relational Model (Preview)

Data stored in tables:

products table:

id	name	price
100	Laptop	1200.00
200	Monitor	300.00

Key Terms

- Table
- Row (record)
- Column (attribute)

What is SQL?

Structured Query Language

Used to:

- Create table
- Query data
- Insert data
- Update data

First SQL Query

```
SELECT 1;
```

👉 SQL can act like a calculator

Create a Table

```
CREATE TABLE products (  
    product_id INTEGER,  
    product_name VARCHAR,  
    price INTEGER  
);
```

Insert Data

```
INSERT INTO products  
VALUES  
(1, 'Laptop', 1000),  
(2, 'Phone-12', 800),  
(3, 'Tablet-1', 500);
```

```
INSERT INTO products (product_id, product_name, price)  
VALUES  
(11, 'Laptop-X', 1200),  
(23, 'Phone-11', 700),  
(35, 'Tablet-2', 500);
```

Query Data

```
SELECT *  
FROM products;
```

Filter Data

```
SELECT *  
FROM products  
WHERE price > 700;
```

Compute Values

```
SELECT name,  
       price,  
       price * 0.9 AS discounted_price  
FROM products;
```

Think Like This

Instead of:

✗ "Write SQL"

Think:

✓ "What question do I want to answer?"

Example Questions

- Which products are expensive?
- Which products are cheap?
- What is the average price?

In-Class Exercise

Ask students:

👉 "Find all products above \$800"

Common Beginner Mistakes

- Forgetting quotes
- Confusing columns vs rows
- Thinking SQL = programming

Mental Model

SQL = Asking questions

Database = Organized memory

Hands-On Lab (Today)

- Run SELECT 1
- Create a table
- Insert data
- Query data
- Filter results

Summary

- Databases are everywhere
- SQL is essential
- Tables are simple structures
- You can already query data 🎉

What's Next?

Week 2:

- Relationships
- Keys
- Data modeling

Final Thought

You are not learning syntax.

👉 You are learning how to think with data.

Let's Practice 